

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1714

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I M. Mohan/19123117 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Mohan

(90)

207 18+23+17+12

60

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CSA1714

Artificial Intelligence

1) Flight delay notification

- propositional logic
- Let D : Flight 202 is delayed.
- Let N : passengers are notified.
- Given : If a flight is delayed, then passengers are notified
- There fore : $D \rightarrow N$
- Fact : D
- Goal : N

2. convert to CNF

• $D \rightarrow N$

$= D \vee N$

• Fact D

3. Negate the goal

• Goal : N

• Negated goal : $\neg N$

4. Resolution

• clause 1 : $D \vee N$

• clause 2 : $\neg N$

• Resolving N and $\neg N$ gives : D

• clause 2 : D

• Resolving D and $\neg D$ gives : $\square$

conclusion

- the empty clause is obtained, the negation of the goal is contradictory.
- There fore, passengers are notified.

2) Employee certification

1. propositional logic

- T : Kevin completed training.

• C = Kevin receives certification.

• Given: $T \rightarrow C$

• Fact: T

• Fact! Goal! C

2. CNF

• $T \rightarrow C$

• $\neg T \vee C$

• Fact: T

3. Negate Goal

• $\neg C$

4. Resolution

• $\neg T \vee C$

• $\neg C$

• Therefore: T

• T and $\neg T$ resolve to $\square$

5. conclusion

• Empty clause = $\square$

• Hence, Kevin receives certification.

3) Medical attention alert.

1. Propositional logic

• F = Ravi has a fever.

• M = Ravi needs medical attention.

• Given: $F \rightarrow M$

• Fact: F

• Goal: M

2. CNF

• $\neg F \vee M$

• F

3. Negate Goal

• $\neg M$

4. Resolution

- $\neg F \vee M$
- $\neg M$
- Therefore: $\neg F$
- F and $\neg F$ resolve to $\square$

5. Conclusion

- Empty clause = $\square$
- Hence, Ravi needs medical attention.

4) Library late fine.

1. Propositional logic

- L = Book was returned late.
- F = Fine is charged.
- Given: $L \rightarrow F$
- Fact: L
- Goal: F

2. CNF

- $L \vee \neg F$
- L

3. Negate Goal

- $\neg F$

4. Resolution

- $L \vee \neg F$
- $\neg F$
- Therefore: L
- L and $\neg L$ resolve to $\square$

Conclusion

- Empty clause = $\square$
- Hence a fine is charged.

5) Vehicle registration process

1. Propositional logic

- E = car passed emissions test.
- C = car is certified roadworthy.
- R = car can be registered

• Given:

- $E \rightarrow C$
- $C \rightarrow R$
- Fact: E

• Goal: R .

2. convert to CNF

- $E \rightarrow C = \neg E \vee C$
- $C \rightarrow R = \neg C \vee R$
- Fact = E

3. Negate goal

- Goal = R
- Negate goal = $\neg R$

4. Resolution steps

- $\neg C \vee R$ and $\neg R \rightarrow \neg C$
- $\neg E \vee C$ and $\neg C \rightarrow \neg E$
- E and $\neg E \rightarrow \square$

5. Conclusion

- Empty clause = $\square$
- Therefore, the car can be registered.